One Equation, Three Scales: $\Sigma = D(\rho_{\text{spacetime}} \| \rho_{\text{matter}})$ as the Universal Action

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[**TO WRITE:** Final abstract after Papers 1–3 are complete.]

Papers 1–3 of this series demonstrated that the entropy production $\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ governs three seemingly unrelated domains: quantum information recovery (Paper 1), strong-field gravity near black holes (Paper 2), and galactic rotation curves (Paper 3). Here we propose that these are not three separate applications but three solutions of *one equation*:

$$\Sigma = D(\rho_{\text{spacetime}} \| \rho_{\text{matter}}),$$

where $\rho_{\text{spacetime}}$ and ρ_{matter} are density matrices encoding the geometry and matter content, respectively, following Bianconi's gravity-from-entropy framework (PRD 2025). Different boundary conditions yield different solutions: strong-field ($r \sim r_s$) $\to$ exponential metric, weak-field ($r \sim$ kpc) $\to$ running G and flat rotation curves, no-gravity limit $\to$ standard quantum information theory. We identify the matching conditions between regimes, derive the emergent arrow of time $\tau = 1 - \exp(-\Sigma/2)$ as a universal consequence, and discuss open problems including the Born rule, Planck-scale physics, and cosmological initial conditions.

THE THREE SCALES

Content outline

[NEW SYNTHESIS] *Summary of Papers 1–3.*—The first three papers of this series established that the entropy production Σ governs physics at three different scales:

TABLE I. The three scales unified by Σ.

	Paper 1	**Paper 2**	**Paper 3**
Domain	QI	Strong field	Weak field
Scale	any	$r \sim r_s$	$r \sim$ kpc
Σ	generic	r_s/r	$r_s/r + \alpha \ln r$
$\mathcal{N}$	any CPTP	grav. redshift	RG-improved grav.
Key result	$F \geq e^{-\Sigma/2}$	exp. metric	flat curves
Status	[KNOWN]	[NEW SYNTHESIS]	[NEW SYNTHESIS]

[NEW SYNTHESIS] *The pattern.*—Each paper uses the *same* mathematical structure:

1. A quantum channel $\mathcal{N}$ that processes information;
2. The QRE drop Σ under $\mathcal{N}$;
3. The Petz recovery map as the optimal retrodiction;
4. The fidelity bound $F \geq \exp(-\Sigma/2)$.

The only difference is the *channel*: a general CPTP map (Paper 1), the gravitational redshift channel (Paper 2), or the RG-improved gravitational channel (Paper 3). This suggests that all three are limits of a single, more fundamental equation.

The $F = ma$ analogy.—Newton's second law $F = ma$ is one equation. With different force laws and boundary conditions:

- Gravity + central field $\to$ Kepler orbits (planets);
- Hooke's law + spring $\to$ simple harmonic oscillation;
- Viscosity + continuum $\to$ Navier–Stokes (fluids).

Nobody says there are "three different $F = ma$'s." There is one equation with three classes of solutions. We propose the same for Σ.

THE UNIFIED EQUATION

Content outline

[SPECULATIVE] *The proposal.*—We conjecture that gravity, quantum information, and the arrow of time are all governed by

$$\boxed{\Sigma = D(\rho_{\text{spacetime}} \| \rho_{\text{matter}}).} \tag{1}$$

Interpretation.—$\rho_{\text{spacetime}}$ encodes the geometry (metric, curvature) as a density matrix. ρ_{matter} encodes the matter content (stress-energy) as a density matrix. The quantum relative entropy between them is the entropy production Σ: the "distance" between what spacetime "is" and what matter "requires" it to be.

[KNOWN] *Formal framework: Bianconi (2025).*—Bianconi [4] provided the mathematical framework in which Eq. (1) is well defined. In the gravity-from-entropy (GfE) framework:

- The metric $g_{\mu\nu}$ is promoted to a density matrix ρ_g on the space of field configurations;
- The matter-induced metric (the metric that matter "wants" via stress-energy) is another density matrix ρ_m;
- The gravitational action is the QRE: $S_{\text{grav}} = D(\rho_g \| \rho_m)$;
- The Einstein equations arise from extremizing $D(\rho_g \| \rho_m)$;
- In the low-coupling limit, this reduces to the standard Einstein–Hilbert action.

[KNOWN] *Bianconi's key results* [4, 5]:

- Modified Einstein equations that reduce to standard GR at low coupling;
- Predicts small positive cosmological constant;
- The "G-field" (extra degree of freedom from QRE structure) is a candidate for dark matter;
- FRW cosmological solutions exist and are thermal [5].

[NEW SYNTHESIS] *Connection to our framework.*—Bianconi's $S_{\text{grav}} = D(\rho_g \| \rho_m)$ is *literally* our $\Sigma = D(\rho_{\text{spacetime}} \| \rho_{\text{matter}})$. The GfE framework provides the formal definition; our series provides the physical interpretation: Σ is the entropy production, $\tau = 1 - \exp(-\Sigma/2)$ is the arrow of time, and the Petz recovery map is the optimal retrodiction.

STRONG-FIELD SOLUTION

Content outline

[NEW SYNTHESIS] *Boundary conditions.*—Static, spherically symmetric, $r \sim r_s$, asymptotically flat.

From Paper 2.—Under these conditions, the entropy production is [2]

$$\Sigma_{\text{strong}} = \frac{r_s}{r} = \frac{2GM}{c^2 r}, \tag{2}$$

and the Petz recovery fidelity gives the metric component

$$g_{00} = -F^2 = -\exp\left(-\frac{r_s}{r}\right). \tag{3}$$

This is the "exponential metric" that replaces Schwarzschild:

- No event horizon (the exponential never reaches zero);
- All curvature invariants remain finite;
- Testable predictions: GW echoes, neutron star redshift, QNM frequencies, shadow size, ISCO radius.

[TO DERIVE: Show that Eq. (2) follows from Eq. (1) under the stated boundary conditions. Specifically: solve $\delta D(\rho_g \| \rho_m) = 0$ for a static, spherically symmetric vacuum with mass M at the origin. The solution should be $g_{00} = -\exp(-r_s/r)$, reproducing Paper 2.]

Matching condition at large r.—As $r \to \infty$, $\Sigma_{\text{strong}} \to 0$ and $g_{00} \to -1$: flat spacetime. The strong-field solution must match onto the weak-field solution (Paper 3) at intermediate radii $r \gg r_s$ but $r \ll r_c \approx 37$ kpc.

WEAK-FIELD SOLUTION

Content outline

[NEW SYNTHESIS] *Boundary conditions.*—Quasi-static, approximately axisymmetric (disk galaxy), $r \sim$ kpc, weak-field ($\Phi/c^2 \ll 1$).

From Paper 3.—Under these conditions, the RG flow of the gravitational channel introduces quantum corrections [3]:

$$\Sigma_{\text{weak}}(r) = \frac{r_s}{r} + \frac{4k_* r_s}{\pi} \ln\left(\frac{r}{r_0}\right), \tag{4}$$

where the logarithmic term arises from the marginal running ($\eta = 1$) of Newton's constant $G(k)$ at IR scales $k \ll k_*$.

Physical consequence.—The logarithmic correction produces an additional $1/r$ force that dominates at $r \gg r_c = 1/k_*$, giving flat rotation curves without dark matter particles.

[TO DERIVE: Show that Eq. (4) follows from Eq. (1) under weak-field boundary conditions with quantum corrections included. This requires: (i) solving $\delta D(\rho_g \| \rho_m) = 0$ at the semiclassical level (reproducing Dorau–Much); (ii) including one-loop corrections via the DPI/RG connection (Casini–Huerta); (iii) showing that the marginal running $\eta = 1$ emerges naturally.]

Matching to strong field.—At $r \gg r_s$ but $r \ll r_c$, the logarithmic correction is negligible and $\Sigma_{\text{weak}} \approx r_s/r$, matching Σ_{strong} smoothly. The two solutions are not disjoint—they are limits of the same equation.

QUANTUM INFORMATION SOLUTION

Content outline

[KNOWN] *Boundary conditions.*—No gravity ($G = 0$), general CPTP map $\mathcal{N}$, finite-dimensional Hilbert spaces.

From Paper 1.—With no gravity, Σ reduces to the generic entropy production of any quantum channel [1]:

$$\Sigma_{\text{QI}} = D(\rho \| \sigma) - D(\mathcal{N}(\rho) \| \mathcal{N}(\sigma)), \tag{5}$$

and the Petz recovery map $\mathcal{R}_{\sigma,\mathcal{N}}$ provides the optimal retrodiction with fidelity bound $F \geq \exp(-\Sigma/2)$.

[NEW SYNTHESIS] *The gravity → QI limit.*—When gravity is turned off, $\rho_{\text{spacetime}} \to \rho_{\text{flat}}$ (Minkowski geometry) and the QRE $D(\rho_{\text{spacetime}} \| \rho_{\text{matter}})$ reduces to the matter-sector entropy production alone. The gravitational contribution vanishes: $\Sigma_{\text{grav}} = 0$, $\tau_{\text{grav}} = 0$. Only the channel-induced entropy production remains.

Connection to quantum error correction.—From Paper 1, $\Sigma = 0 \Leftrightarrow \tau = 0 \Leftrightarrow$ perfect Petz recovery $\Leftrightarrow$ quantum Markov chain $\Leftrightarrow$ Knill–Laflamme conditions. QEC is the deliberate construction of a subsystem where $\Sigma = 0$: an artificial elimination of the arrow of time within a local code space.

MATCHING CONDITIONS AND PREDICTIONS

Content outline

[NEW SYNTHESIS] *The three regimes are connected.*—The solutions of Eq. (1) in different boundary conditions must *match* at their boundaries:

1. *Strong → weak*: At $r \gg r_s$, the exponential metric linearizes to Newton + small corrections. The logarithmic correction from Paper 3 emerges as the leading quantum correction to this classical limit.
2. *Weak → QI*: At scales where $\Phi/c^2 \to 0$ (laboratory), the gravitational Σ becomes negligible and only quantum-channel entropy production remains.
3. *Weak → cosmological*: At $r \sim r_H = c/H_0$ (Hubble radius), the de Sitter background contribution $\Sigma_{\text{dS}} \sim r/r_H$ becomes dominant, connecting galactic dynamics to cosmic expansion.

[SPECULATIVE] *Intermediate-scale predictions.*—At scales where two regimes overlap, *both* corrections matter simultaneously. Specific predictions:

- At $r \sim 100$ kpc (galaxy cluster outskirts): both running-G and de Sitter corrections contribute. This may resolve the Bullet Cluster puzzle if the two effects combine to produce the observed mass-light separation.
- At $r \sim r_s \cdot k_*/k_s$ (strong-field with quantum corrections): the exponential metric acquires logarithmic modifications. Testable via precision measurements of neutron star properties.

[SPECULATIVE] *The three-term approximation.*—In the appropriate limits, the unified Σ can be approximated as

$$\Sigma(r) \approx \underbrace{\frac{r_s}{r}}_{\text{local gravity}} + \underbrace{\alpha \ln\left(\frac{r}{r_c}\right)}_{\text{RG running}} + \underbrace{\beta \frac{r}{L_{\text{dS}}}}_{\text{de Sitter}}, \tag{6}$$

where:

- Term 1 dominates near compact objects ($r \lesssim r_s$);
- Term 2 dominates at galactic scales ($r_c \lesssim r \lesssim L_{\text{dS}}$);
- Term 3 dominates at cosmological scales ($r \sim L_{\text{dS}}$).

This is an approximation, not the fundamental equation. The fundamental equation is Eq. (1); the three terms emerge as limits.

THE ARROW OF TIME AS EMERGENT

Content outline

[NEW SYNTHESIS] *The central thesis.*—The arrow of time is not a separate law of nature. It *is* the entropy production Σ:

$$\tau = 1 - \exp(-\Sigma/2). \tag{7}$$

This single equation, applied across scales, gives:

TABLE II. The arrow of time at different scales.

Scale	Σ	τ	**Physical meaning**
Closed system	0	0	No arrow of time
Open (lab)	> 0	> 0	Arrow exists
Near BH	r_s/r	$\to 1$	Strong arrow
Galaxy edge	$+\alpha \ln r$	> 0	Arrow from RG
Hubble horizon	~ 1	~ 0.39	Cosmic arrow
Singularity	$\to \infty$	$= 1$	Maximal arrow

[NEW SYNTHESIS] *Three key insights.*—

1. $\tau = 0$ *(closed system): time does not exist.*—When $\Sigma = 0$, perfect Petz recovery is possible: the "future" state can be perfectly retrodicted from the "past" state, and vice versa. Past and future are operationally indistinguishable. This is not a philosophical statement but an operational one: no measurement can determine the temporal ordering of events.
2. $\tau > 0$ *(open system): time's arrow exists.*—When $\Sigma > 0$, retrodiction is imperfect: there is a preferred direction. The degree of preference is continuously quantified by $\tau \in (0, 1)$.
3. $\tau \to 1$ *(macroscopic/singularity): classical world.*—When $\tau \approx 1$, the arrow of time is absolute. Collapse is complete. Classical physics emerges. This is the everyday world.

[NEW SYNTHESIS] *The signed version.*—The full picture requires

$$\tau_{\text{signed}} = \ln\left(\frac{P_{\text{fwd}}}{P_{\text{rev}}}\right), \tag{8}$$

which allows three cases:

- $\tau_{\text{signed}} > 0$: forward arrow (normal);
- $\tau_{\text{signed}} = 0$: no arrow (closed/unitary);
- $\tau_{\text{signed}} < 0$: reverse arrow (non-Markovian backflow, spin echo, Wang et al. 2002 colloidal particle experiments [20]).

The second law ($\langle\Sigma\rangle \geq 0$) is a *statistical* law: $P(\Sigma < 0)/P(\Sigma > 0) = \exp(-|\Sigma|)$ (Crooks). Time reversal is not impossible, merely exponentially unlikely for macroscopic systems.

COSMOLOGICAL EXTENSION

Content outline

[SPECULATIVE] *FRW from* Σ.—In the Friedmann–Robertson–Walker (FRW) universe, there is no timelike Killing vector. The natural replacement is the Kodama vector K^a [14], which:

- Is divergence-free in any spherically symmetric spacetime;
- Gives a conserved charge = Misner–Sharp mass;
- Defines surface gravity at the apparent horizon.

The cosmological τ can be defined via [34]:

$$F_K = \frac{|K|}{|K_{\text{ref}}|}, \tag{9}$$

giving $\tau = 0$ at the observer ($r = 0$) and $\tau = 1$ at the apparent horizon ($|K|^2 = 0$).

[SPECULATIVE] *Friedmann equations from QRE.*—The first law of thermodynamics at the FRW apparent horizon, $dE = TdS + WdV$, projects along the Kodama vector to give the second Friedmann equation [15, 16]. Combined with Dorau–Much and Bianconi, this suggests:

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho \quad \Leftarrow \quad \delta\, D(\rho_{\text{st}}\|\rho_{\text{m}}) = 0. \tag{10}$$

[**TO DERIVE:** Derive the Friedmann equations from $\Sigma = D(\rho_{\text{spacetime}}\|\rho_{\text{matter}})$ using Bianconi's GfE framework with FRW boundary conditions. Bianconi [5] has shown FRW solutions exist; the task is to reproduce the standard Friedmann equations as a limit.]

[SPECULATIVE] *Dark energy from finite cosmic information.*— Padmanabhan's CosMIn principle [17]: the total cosmic information I_{CosMIn} transferred from quantum to classical must be *finite*. This requires late-time accelerated expansion ($\Lambda > 0$). In the τ language:

$$I_{\text{CosMIn}} = \int_0^\infty \tau(t)\, dt < \infty \quad \Rightarrow \quad \Lambda > 0. \tag{11}$$

[**TO WRITE:** Paragraph on the interpretation: $\Lambda > 0$ is not a "mystery" but a *necessity* for finite cosmic information processing. The universe must eventually stop producing information (entropy) at an unbounded rate. Accelerated expansion achieves this by diluting matter and radiation.]

THE ANALOGY MADE PRECISE

Content outline

[NEW SYNTHESIS] *One equation, three solutions.*—We now make the $F = ma$ analogy precise:

What the analogy captures.—

- Just as $F = ma$ does not tell you the force law (you need to specify gravity, Hooke, etc.), $\Sigma = D(\rho_{\text{st}}\|\rho_{\text{m}})$ does not tell you the channel (you need to specify the boundary conditions).
- Just as different solutions of $F = ma$ look completely different (ellipses vs. oscillations vs. turbulence), different solutions of the unified Σ look completely different (black holes vs. rotation curves vs. quantum channels).
- Just as $F = ma$ has a deeper formulation (Lagrangian mechanics, Hamilton's principle), the unified Σ has a deeper formulation via the Petz recovery map as the canonical retrodiction functor [10].

OPEN PROBLEMS

Content outline

[**TO WRITE:** Each open problem should be 2–3 paragraphs: statement, current status, possible approach.]

Problem 1: First-principles derivation.—Derive $\Sigma = D(\rho_{\text{spacetime}}\|\rho_{\text{matter}})$ from more primitive axioms. Currently, Eq. (1) is a *conjecture* motivated by Bianconi's GfE [4], Dorau–Much's QRE $\to$ Einstein equations [6], and the consistency of Papers 1–3. A derivation from axioms (e.g., von Neumann algebra axioms, or the axioms of algebraic QFT) would elevate it from conjecture to theorem.

TABLE III. The $F = ma$ analogy for $\Sigma = D(\rho_{\mathrm{st}}\|\rho_{\mathrm{m}})$.

	$F = ma$	$\Sigma = D(\rho_{\mathrm{st}}\|\rho_{\mathrm{m}})$	**Status**
Equation	$\mathbf{F} = m\mathbf{a}$	$\Sigma = D(\rho_{\mathrm{st}}\|\rho_{\mathrm{m}})$	[SPECULATIVE]
Solution 1	Kepler orbits (gravity, $r \gg 0$)	Exponential metric (strong field, $r \sim r_s$)	[NEW SYNTHESIS]
Solution 2	SHM (Hooke, spring)	Flat rotation curves (weak field, $r \sim$ kpc)	[NEW SYNTHESIS]
Solution 3	Navier–Stokes (continuum)	Standard QI ($G = 0$, any CPTP)	[KNOWN]
Solution 4	—	FRW cosmology ($r \sim L_H$)	[SPECULATIVE]
Matching	Virial theorem	$\Sigma_{\mathrm{strong}} \to \Sigma_{\mathrm{weak}}$ at $r \gg r_s$	**[TO DERIVE:]**
Deep structure	Lagrangian mechanics	Petz recovery map as retrodiction functor	[KNOWN]

[SPECULATIVE] Current best candidates: (i) Bianconi's entropic action principle [4]; (ii) Dorau–Much's modular theory [6] extended to include backreaction; (iii) Jacobson's entanglement equilibrium [12] rewritten in QRE language.

Problem 2: Planck scale.—What happens when spacetime itself is quantum? At the Planck scale, $\rho_{\mathrm{spacetime}}$ is not a semiclassical density matrix but a fully quantum object. The QRE $D(\rho_{\mathrm{spacetime}}\|\rho_{\mathrm{matter}})$ must be replaced by an algebraic QFT version using the Araki relative entropy [13]. This is mathematically well defined but computationally intractable with current methods.

Problem 3: Cosmological initial conditions.—Why does the universe start with $\Sigma \approx 0$ (low entropy)? In the τ framework, this is the question: why does $\tau_{\mathrm{initial}} \approx 0$? One possibility: the initial state is a "Petz-recovered" state from the final condition (a closed-universe scenario where boundary conditions at both ends enforce $\Sigma = 0$). This is highly speculative and connects to the "past hypothesis" of thermodynamics [22, 23].

Problem 4: The Born rule.—The τ framework assumes the Born rule in computing fidelity F and entropy production Σ. A complete theory should derive the Born rule, not assume it. Possible connections:

- Zurek's "envariance" derivation [21] uses environment-assisted symmetry, which connects to τ via quantum Darwinism;
- Deutsch–Wallace decision-theoretic approach relies on rational betting, which maps to Petz recovery optimality;
- Gleason's theorem constrains probability measures on Hilbert space, and the Petz map is the unique "probability measure on channels."

[SPECULATIVE] *Problem 5: Consciousness.*—We acknowledge but do not attempt this problem. The hard problem of consciousness (why subjective experience exists) is sometimes connected to the arrow of time and measurement. The τ framework provides a precise, quantitative arrow of time that does not require an observer. Whether this has implications for consciousness is a question for philosophers, not for this paper.

Problem 6: Derive $\eta = 1$.—The marginal anomalous dimension $\eta = 1$ that produces flat rotation curves (Paper 3) should follow from the properties of $D(\rho_{\mathrm{spacetime}}\|\rho_{\mathrm{matter}})$ in the IR. This is the most concrete open problem and the one most likely to be resolved in the near term. The DPI structure of QRE may constrain η to exactly 1 via a saturation argument.

Problem 7: Gravitational waves and dynamical spacetimes.— Papers 2 and 3 address static or quasi-static spacetimes. Extending to dynamical spacetimes (gravitational wave emission, binary mergers) requires:

- A time-dependent $\Sigma(t)$;
- The Kodama vector (for spherical symmetry) or its generalization [31];
- Raychaudhuri decomposition of $d\Sigma/ds$ along geodesic congruences.

CONCLUSION

Content outline

[TO WRITE: Concluding section synthesizing the full four-paper arc.]

Summary.—We have argued that a single equation, $\Sigma = D(\rho_{\mathrm{spacetime}}\|\rho_{\mathrm{matter}})$, unifies three scales of physics:

- Quantum information (Paper 1): $\Sigma = 0 \Leftrightarrow$ perfect recovery, no arrow of time;

- Strong-field gravity (Paper 2): $\Sigma = r_s/r \to$ exponential metric, information-theoretic "soft horizon";
- Galactic dynamics (Paper 3): $\Sigma = r_s/r + \alpha \ln r \to$ running G, flat rotation curves without dark matter particles.

The philosophical point.—

> We have not discovered a new equation. We have proposed that an existing mathematical structure—quantum relative entropy and its recovery theory—when applied consistently across scales, unifies phenomena that appear unrelated. The contribution is not a new formula but a new viewpoint that makes existing results cohere. This is the kind of contribution that, if correct, simplifies rather than complicates.

[TO WRITE: Final paragraph on falsifiability: the framework makes specific, testable predictions at each scale (Paper 2: GW echoes, shadow sizes; Paper 3: universal r_c, SPARC fits, BIG-SPARC). If these predictions fail, the framework fails. This is a feature, not a bug.]

The author thanks the quantum information, quantum gravity, astrophysics, and quantum foundations communities for the foundational works on which this synthesis builds.

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